Roll No.:

ODD MID TERM EXAMINATION Sept 2025

Name of the Program: ***M. Tech (Electric Vehicle Technology)*** Semester: 1st
Name of the Course: Automotive Technology Course Code: MEV 103
Time: 90 Minutes

Note:

i. Answer all the questions by choosing any one of the sub-questions.
ii. Each question carry 10 marks

Q1	(10 marks)	CO1
(a)	List the main components of an automobile and discuss their functions	
	OR	
(b)	Define and classify the chassis. Discuss its functions.	
Q2	(10 marks)	CO1
(a)	Explain front engine front wheel drive with neat sketch	
	OR	
(b)	Explain in detail the principle of Four-wheel drive with a sketch.	
Q3		CO1
(a)	Define aerodynamics. Explain air resistance, rolling resistance, and gradient resistance acting on a vehicle.	
	OR	
(b)	Explain the construction and function of the piston, connecting rod, and crankshaft in an I.C. engine with neat sketches.	
Q4	(10 marks)	CO2
(a)	Explain the principle of the valve timing diagram for a four-stroke petrol engine. Discuss the significance of valve overlap.	
	OR	
(b)	Explain the Electronic Controlled Gasoline Injection System (ECGIS) with its three main subsystems. Draw a neat block diagram.	
Q5	(10 marks)	CO2
(a)	Write a short note on the Common Rail Direct Injection (CRDI) system and its advantages over conventional diesel systems.	
	OR	
(b)	Explain Multi Point Fuel Injection (MPFI) system with a neat sketch with its advantages and disadvantages.	